SCSI 1013 DISCRETE STRUCTURE



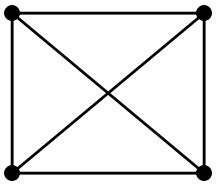
CHAPTER 4 – PART 2

TREE

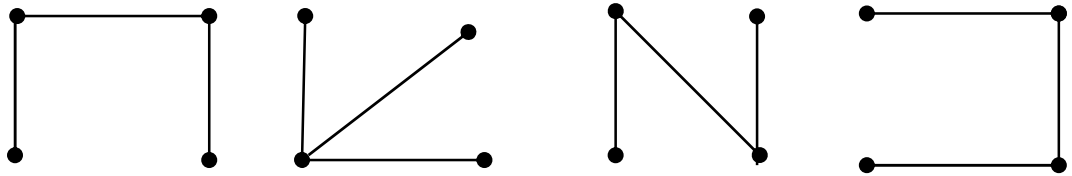
Semester 1 – 2017/2018

Spanning Trees

- A spanning tree is a simple graph that is a subgraph of ***G*** and contains every vertex of ***G*** and is a tree.



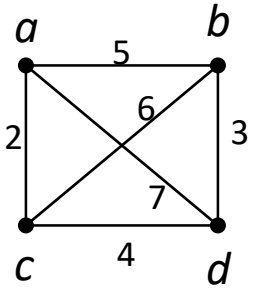
A connected undirected graph

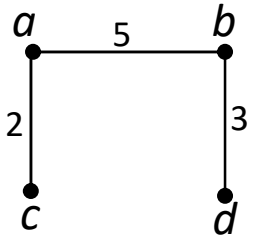


Four spanning trees of the graph

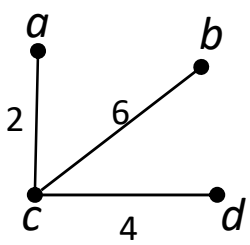
Minimum Spanning Tree (MST)

- A Minimum Spanning Tree is a spanning tree on a weighted graph that has minimum total weight.
- Example

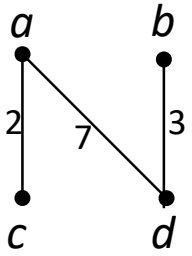




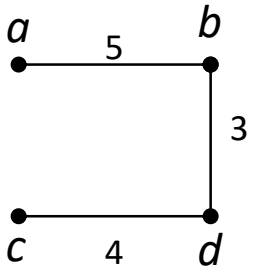
$T_1 = 10$



$T_2 = 12$

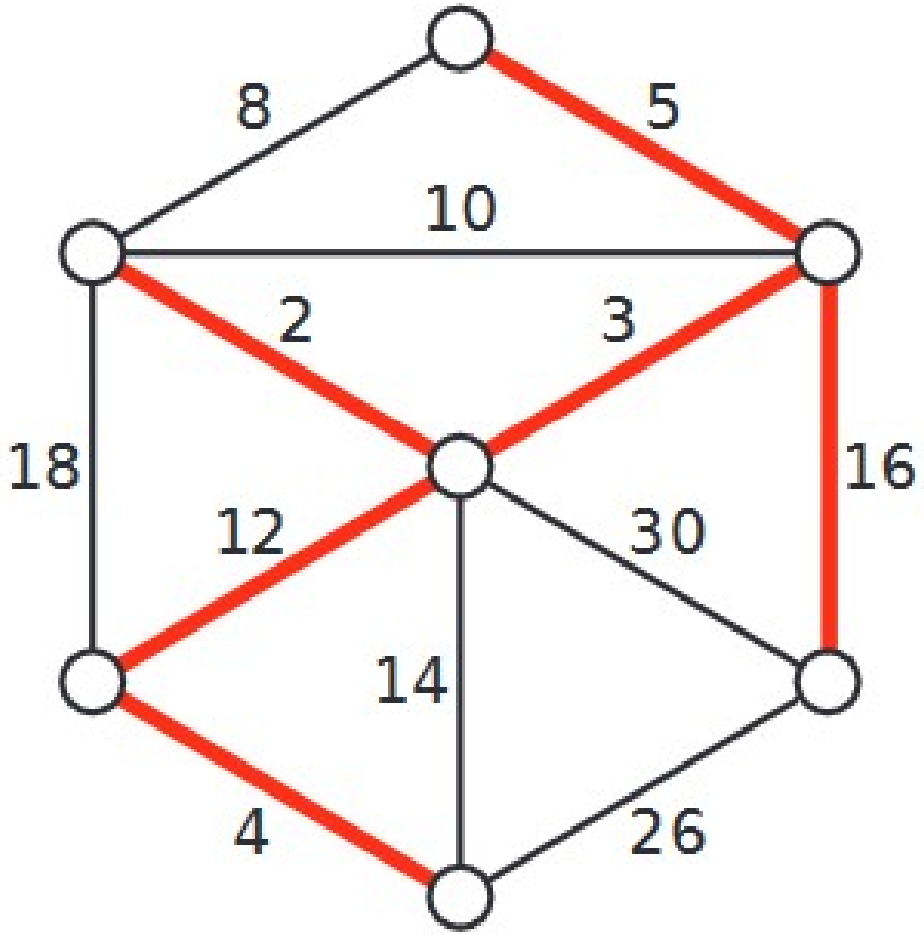


$T_3 = 12$



$T_4 = 12$

Minimum Spanning Tree (MST)



A weighted graph and its minimum spanning tree.

Muddy City Problem

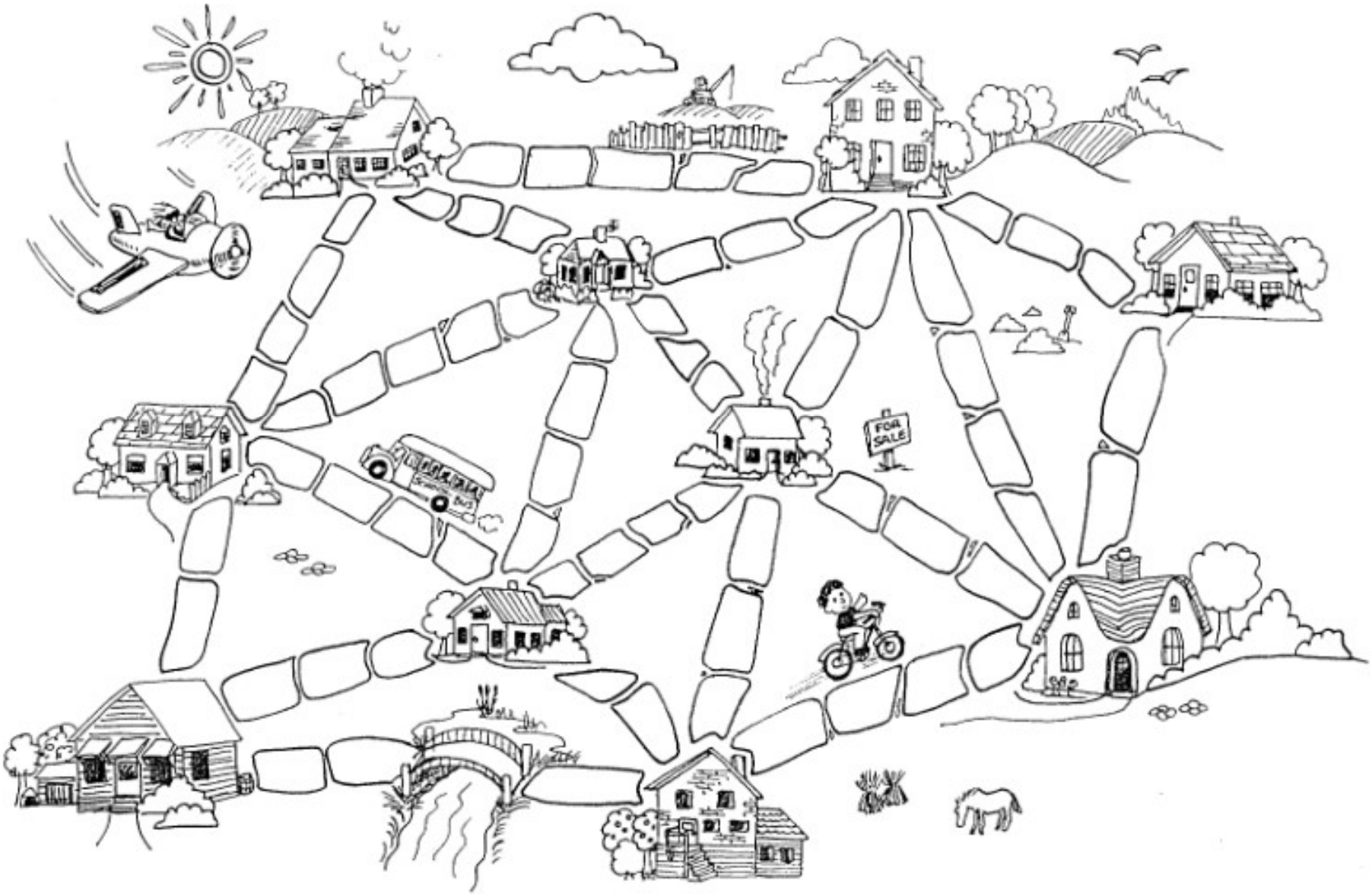
Once upon a time there was a city that had no roads. Getting around the city was particularly difficult after rainstorms because the ground became very muddy. Cars got stuck in the mud and people got their boots dirty. The mayor of the city decided that some of the streets must be paved, but didn't want to spend more money than necessary because the city also wanted to build a swimming pool.

Muddy City Problem

The mayor therefore specified two conditions:

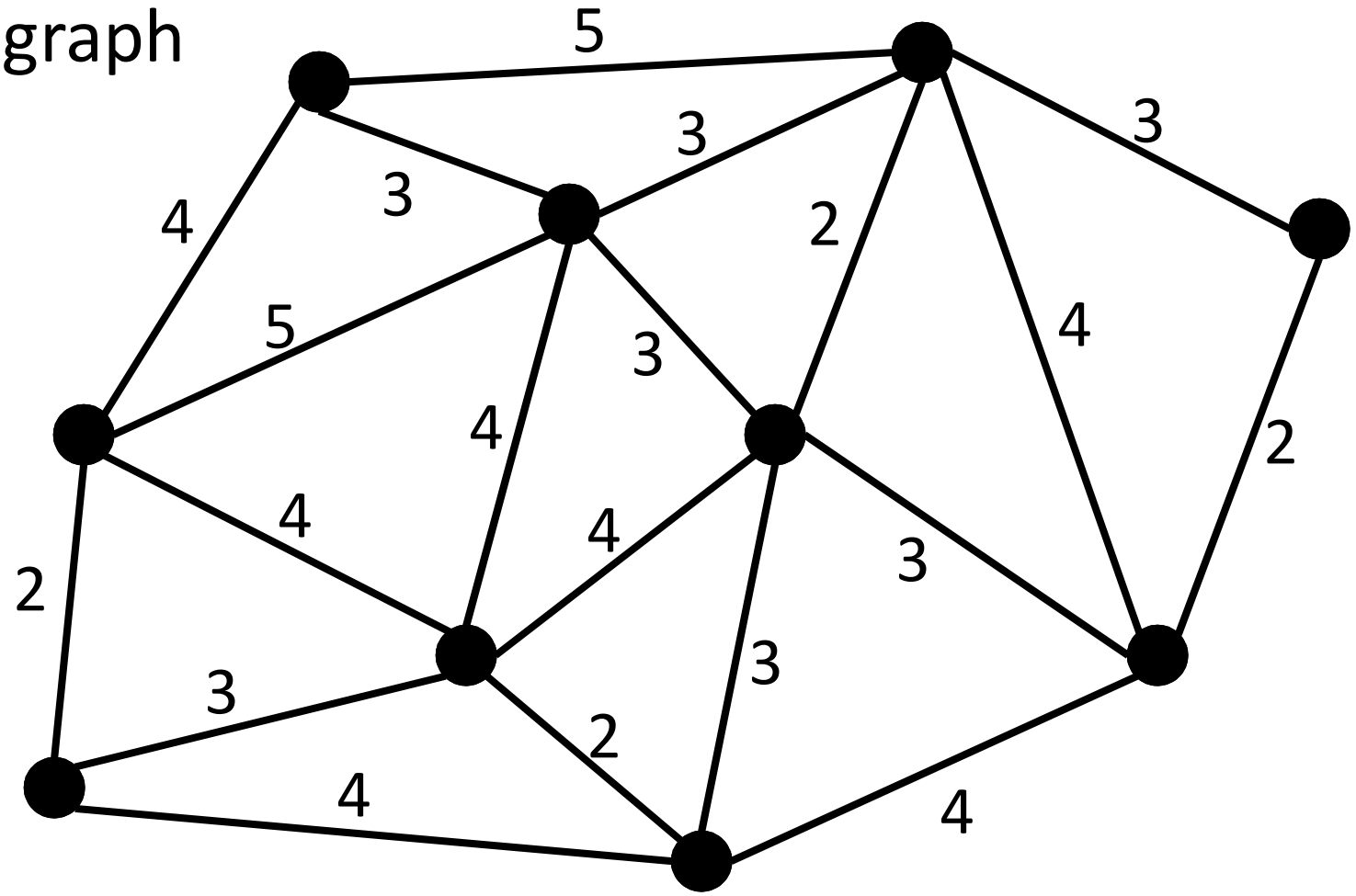
1. Enough streets must be paved so that it is possible for everyone to travel from their house to anyone else's house only along paved roads, and
2. The paving should cost as little as possible.

Here is the layout of the city. The number of paving stones between each house represents the cost of paving that route. Find the best route that connects all the houses, but uses as few counters (paving stones) as possible.


FOR SALE

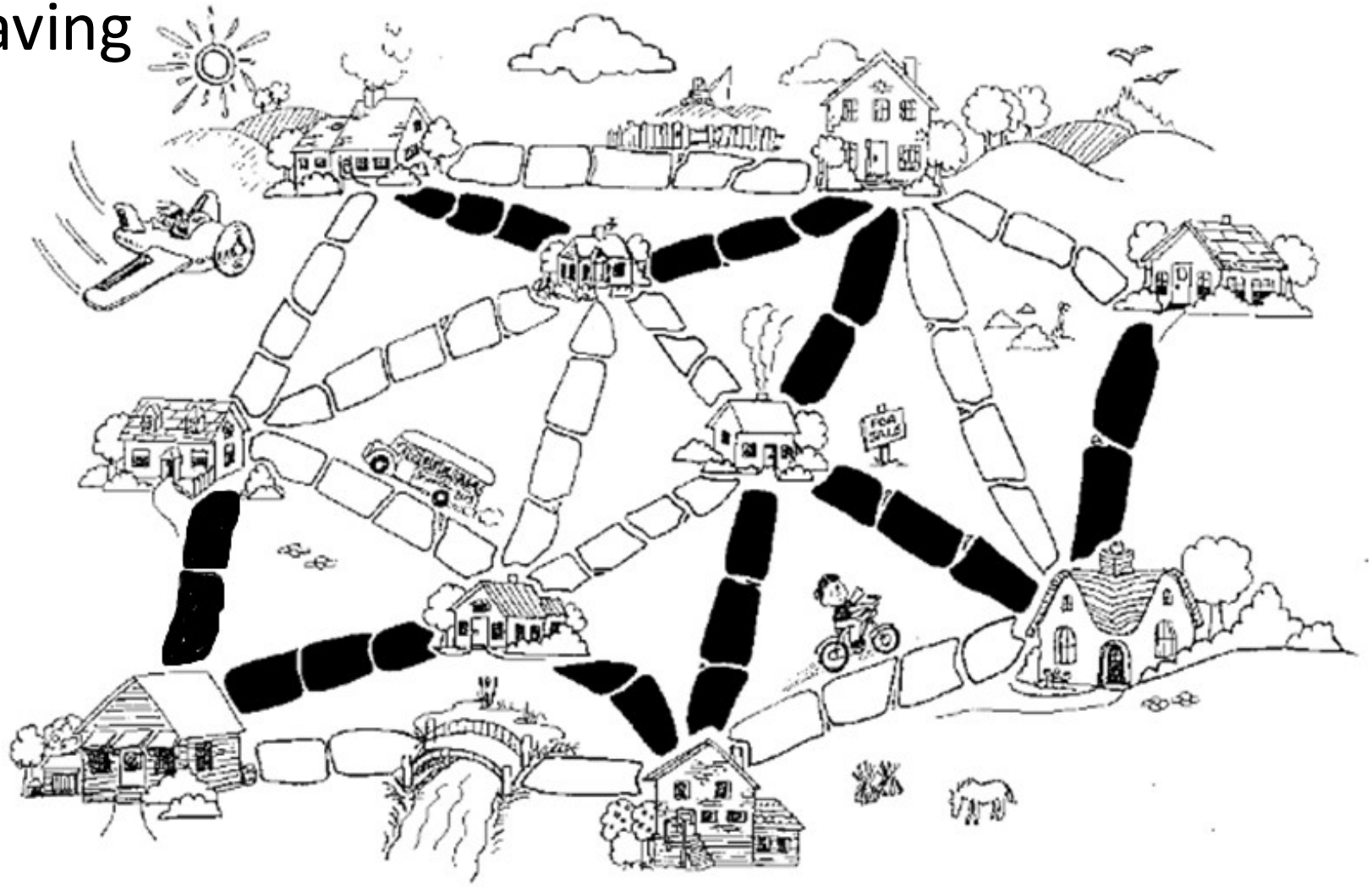
Muddy City Problem

The graph



Muddy City Problem

The paving



Application of MST: Examples

- In the design of electronic circuitry, it is often necessary to make a set of pins electrically equivalent by wiring them together.
- Running cable TV to a set of houses. What's the least amount of cable needed to still connect all the houses?

Finding MST

- Kruskal's algorithm: start with no nodes or edges in the spanning tree and repeatedly add the cheapest edge that does not create a cycle

Kruskal algorithm

Procedure Kruskal (*G*: weighted connected undirected graph with *n* vertices)

T:= empty graph

for i := 1 to *n*-1

begin

e:= any edge in *G* with smallest weight that does not

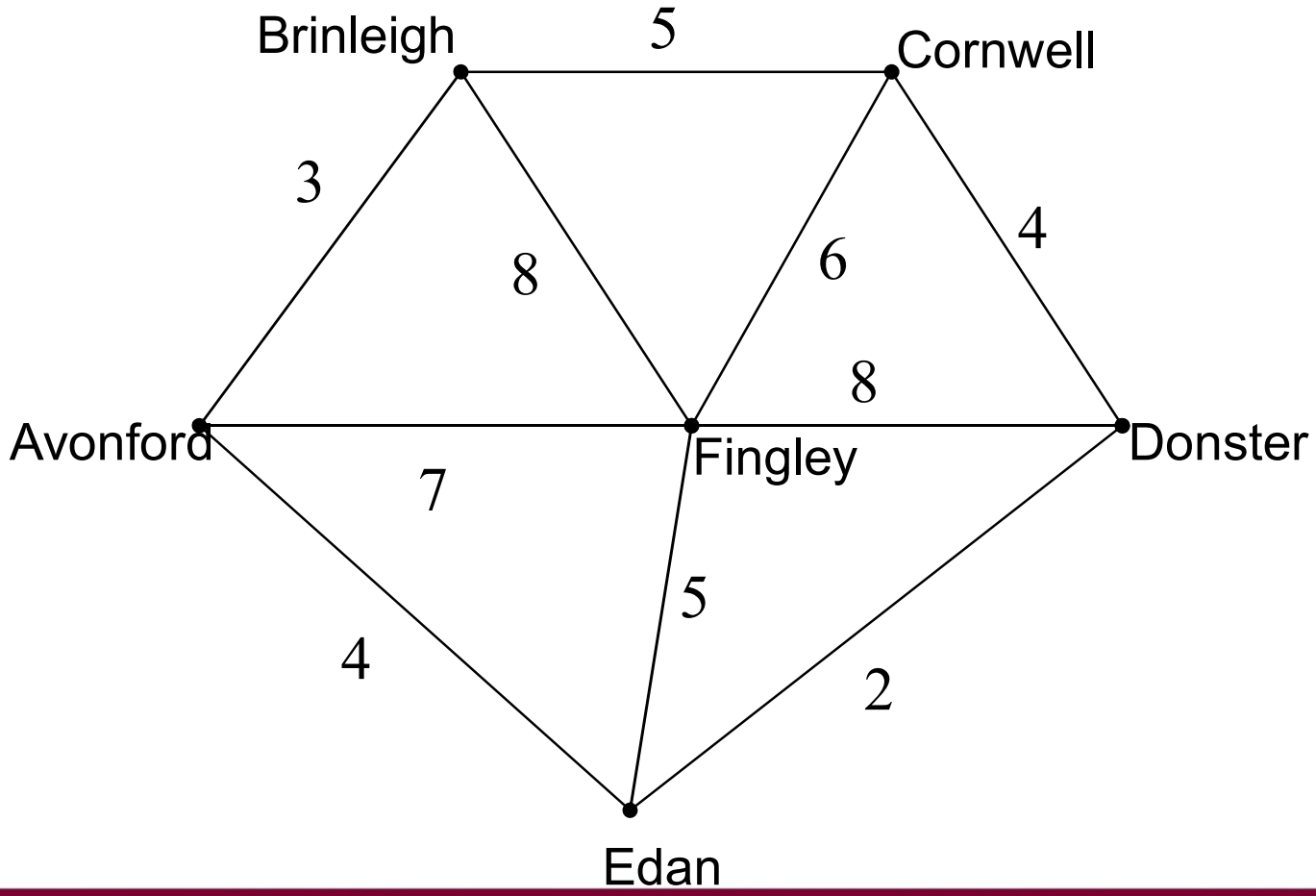
form a simple circuit when added to *T*

T:= *T* with e added

end (*T* is a minimum spanning tree of *G*)

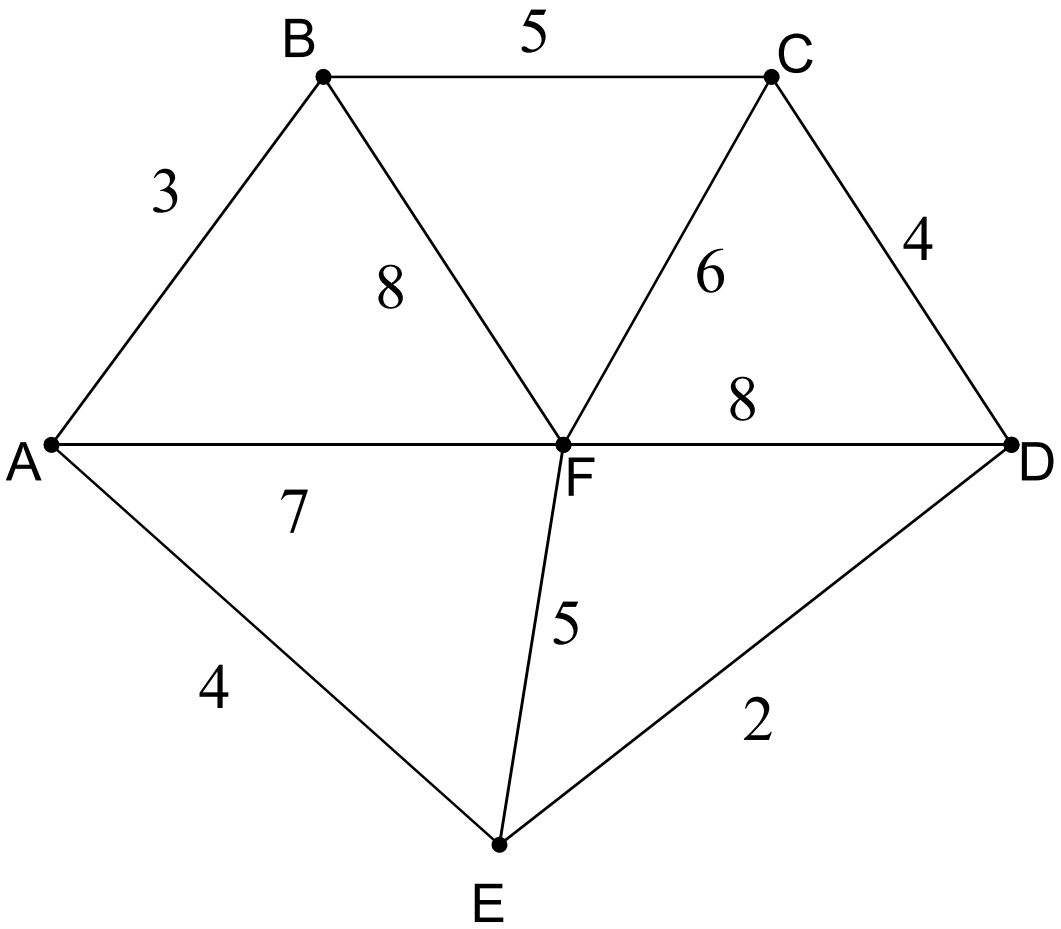
Example

A cable company want to connect five villages to their network which currently extends to the market town of Avonford. What is the minimum length of cable needed?

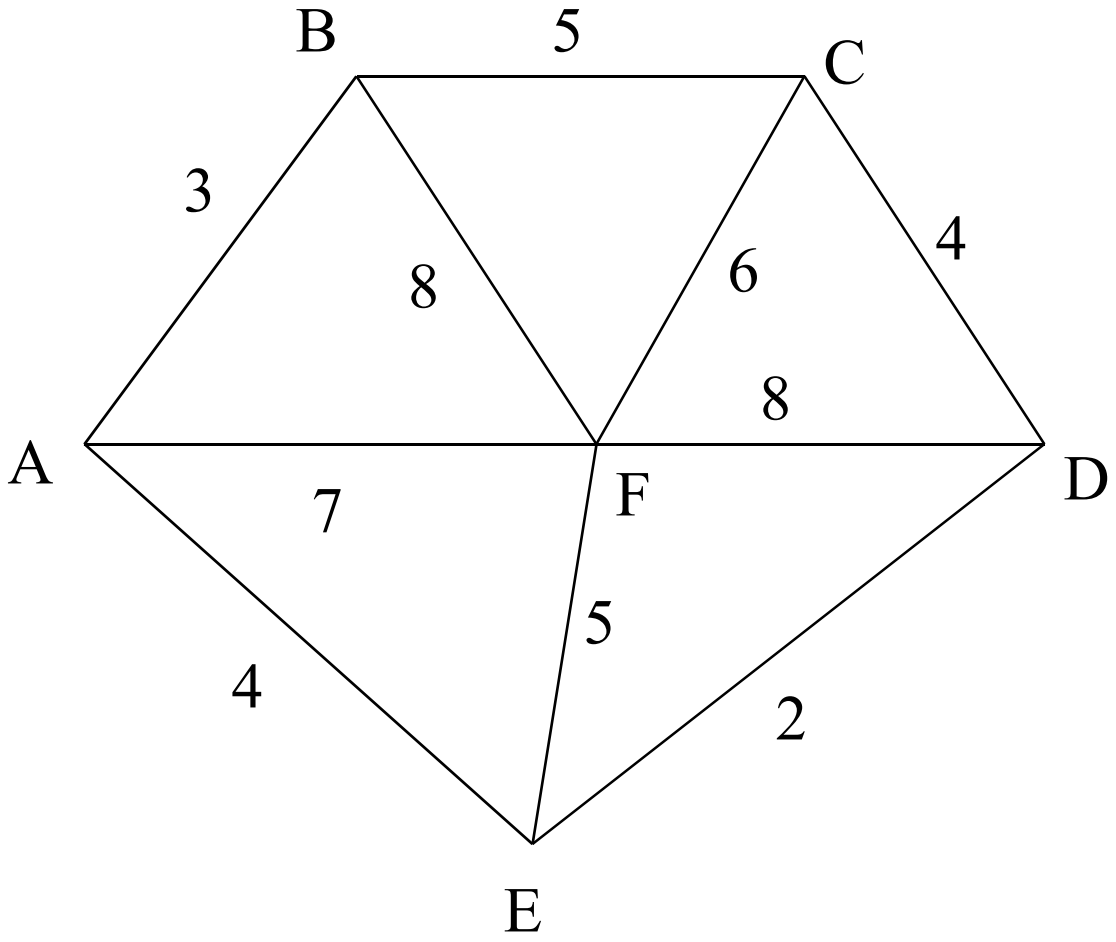


Example

We model the situation as a network, then the problem is to find the minimum connector for the network.



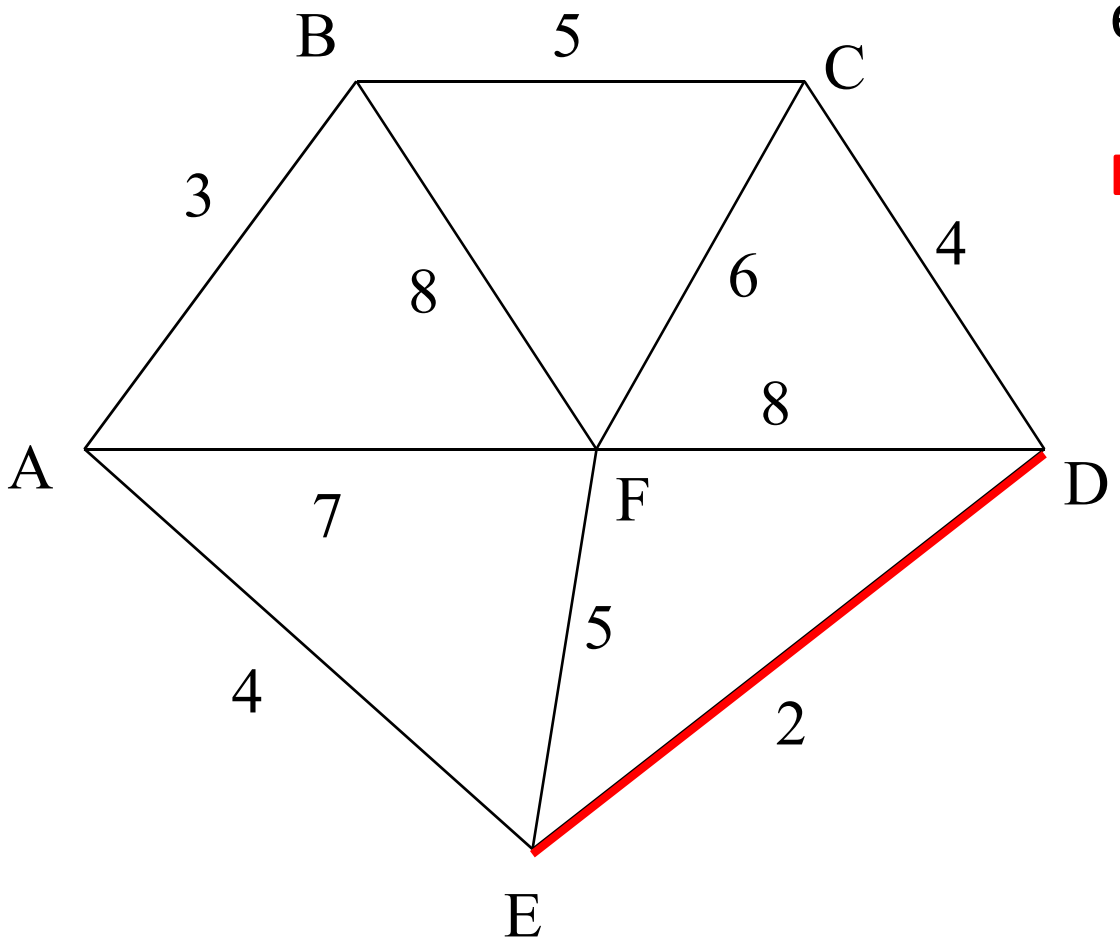
Kruskal's Algorithm



List the edges in order of size:

ED	2
AB	3
AE	4
CD	4
BC	5
EF	5
CF	6
AF	7
BF	8
CF	8

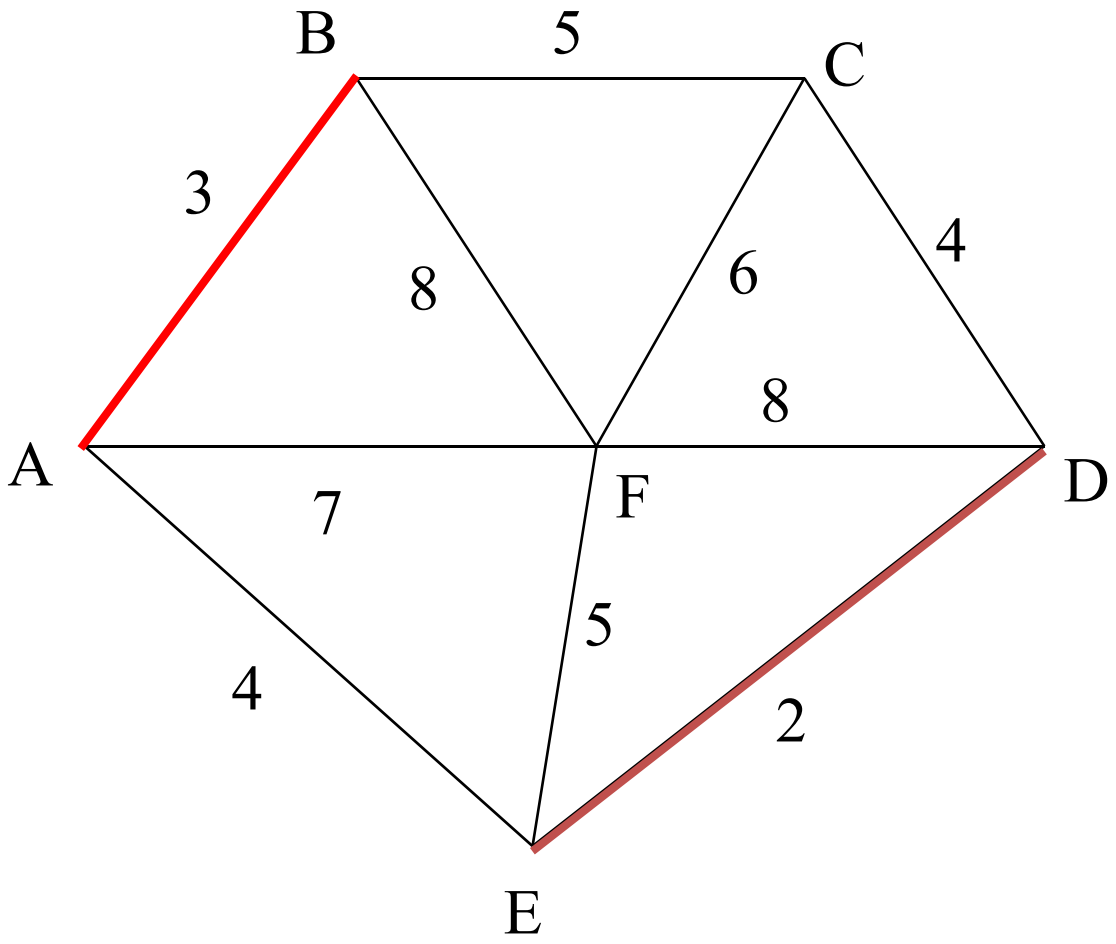
Kruskal's Algorithm



Select the shortest edge in the network

ED 2

Kruskal's Algorithm

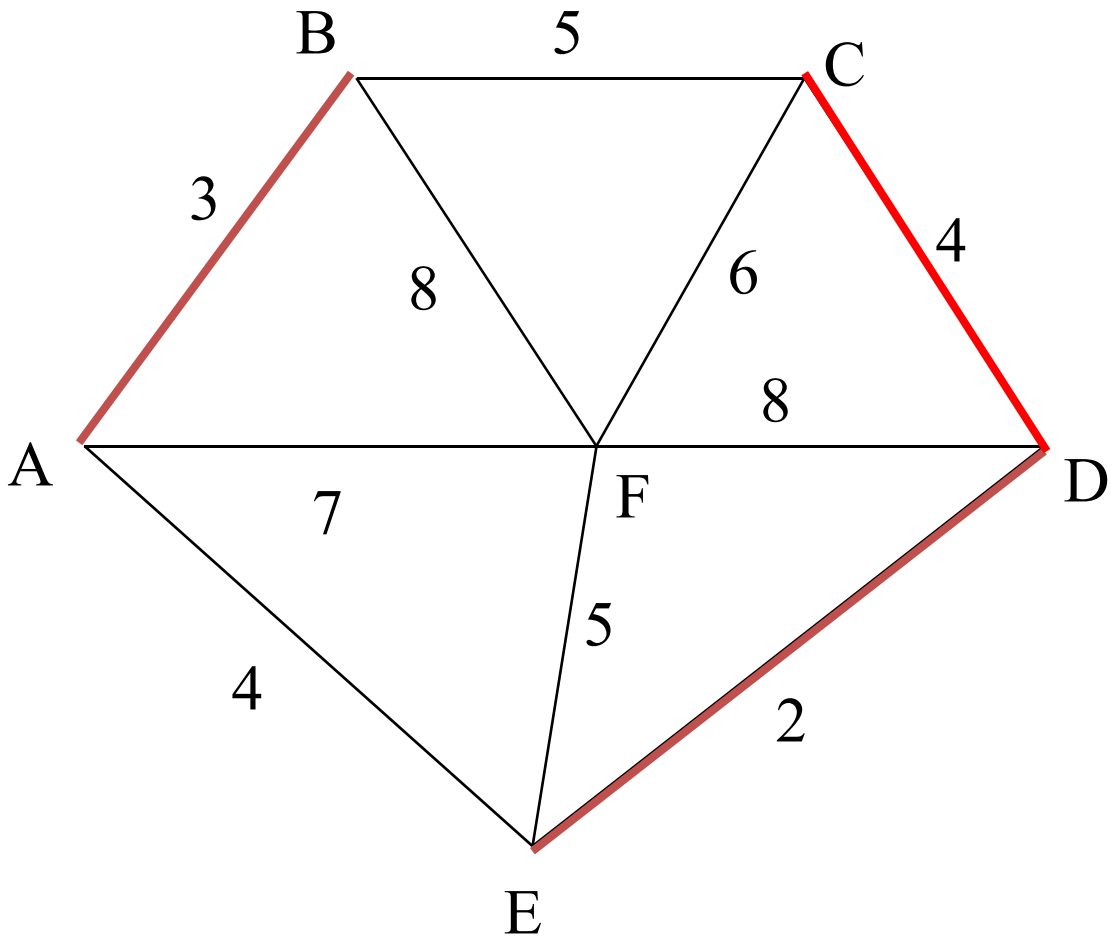


Select the next shortest edge which does not create a cycle

ED 2

AB 3

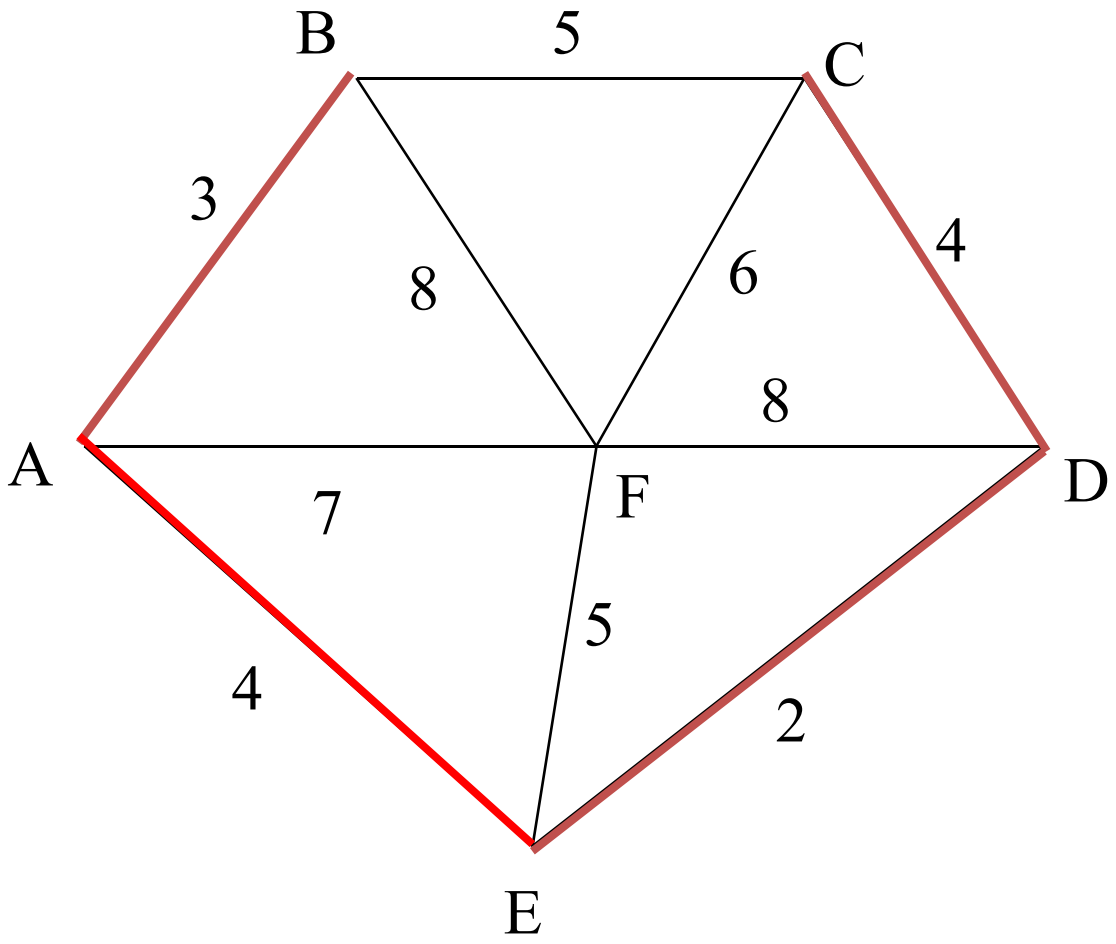
Kruskal's Algorithm



Select the next shortest edge which does not create a cycle

ED 2
AB 3
CD 4 (or AE 4)

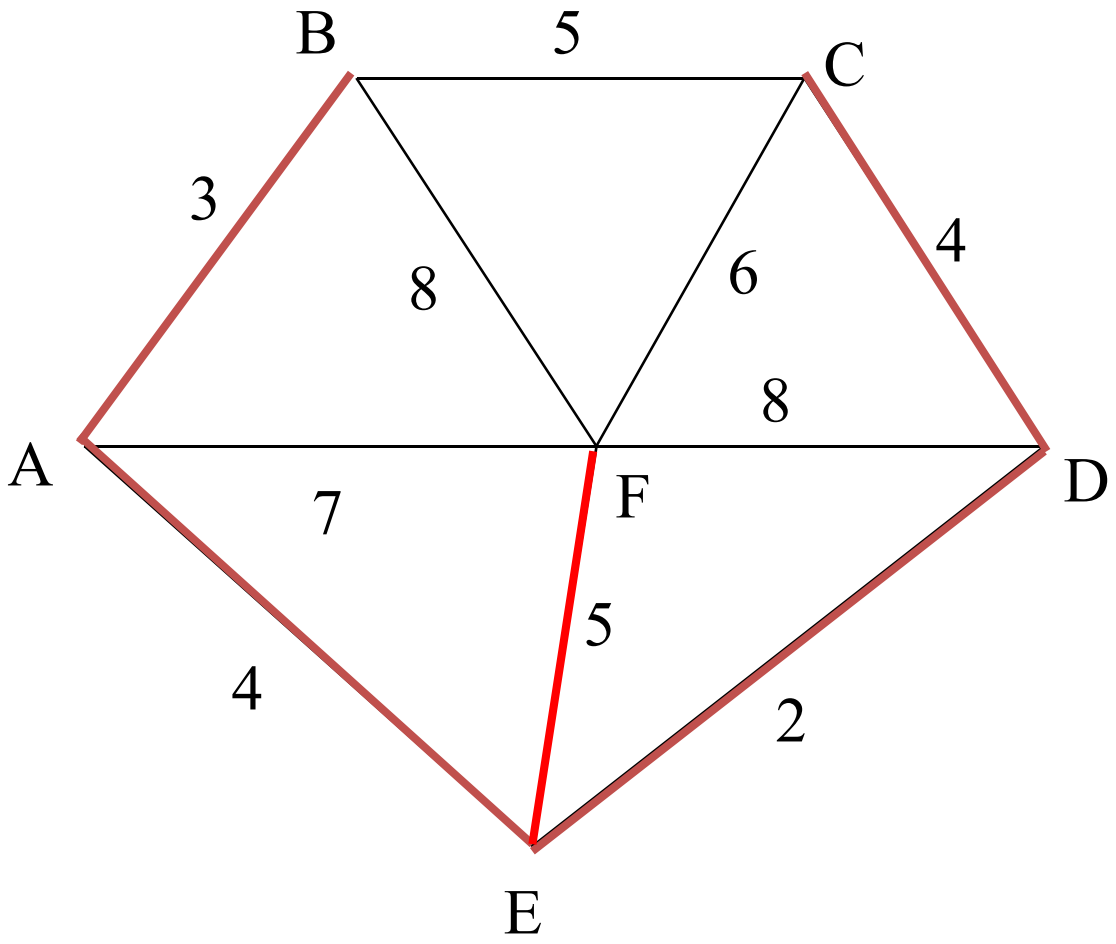
Kruskal's Algorithm



Select the next shortest edge which does not create a cycle

ED 2
AB 3
CD 4
AE 4

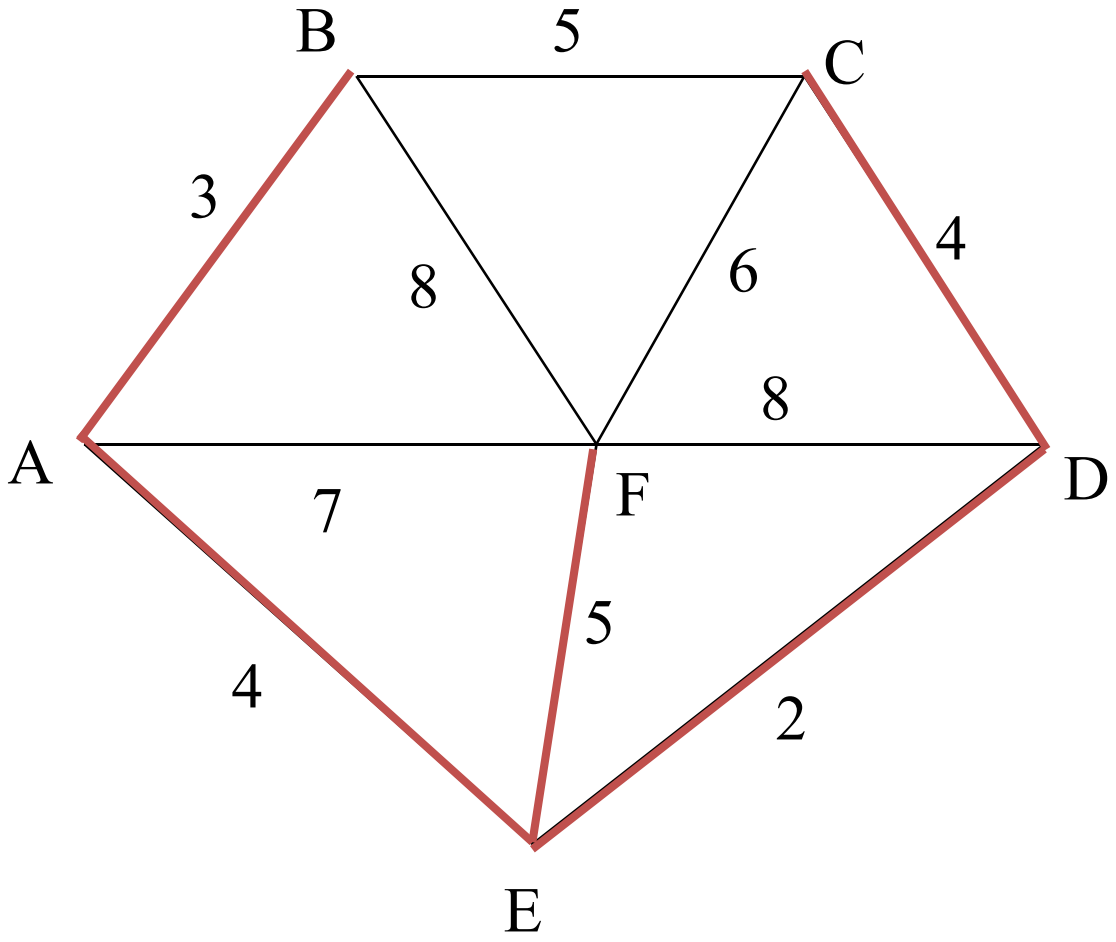
Kruskal's Algorithm



Select the next shortest edge which does not create a cycle

ED 2
AB 3
CD 4
AE 4
~~BC 5~~ – forms a cycle
EF 5

Kruskal's Algorithm



All vertices have been connected.

The solution is

ED 2
AB 3
CD 4
AE 4
EF 5

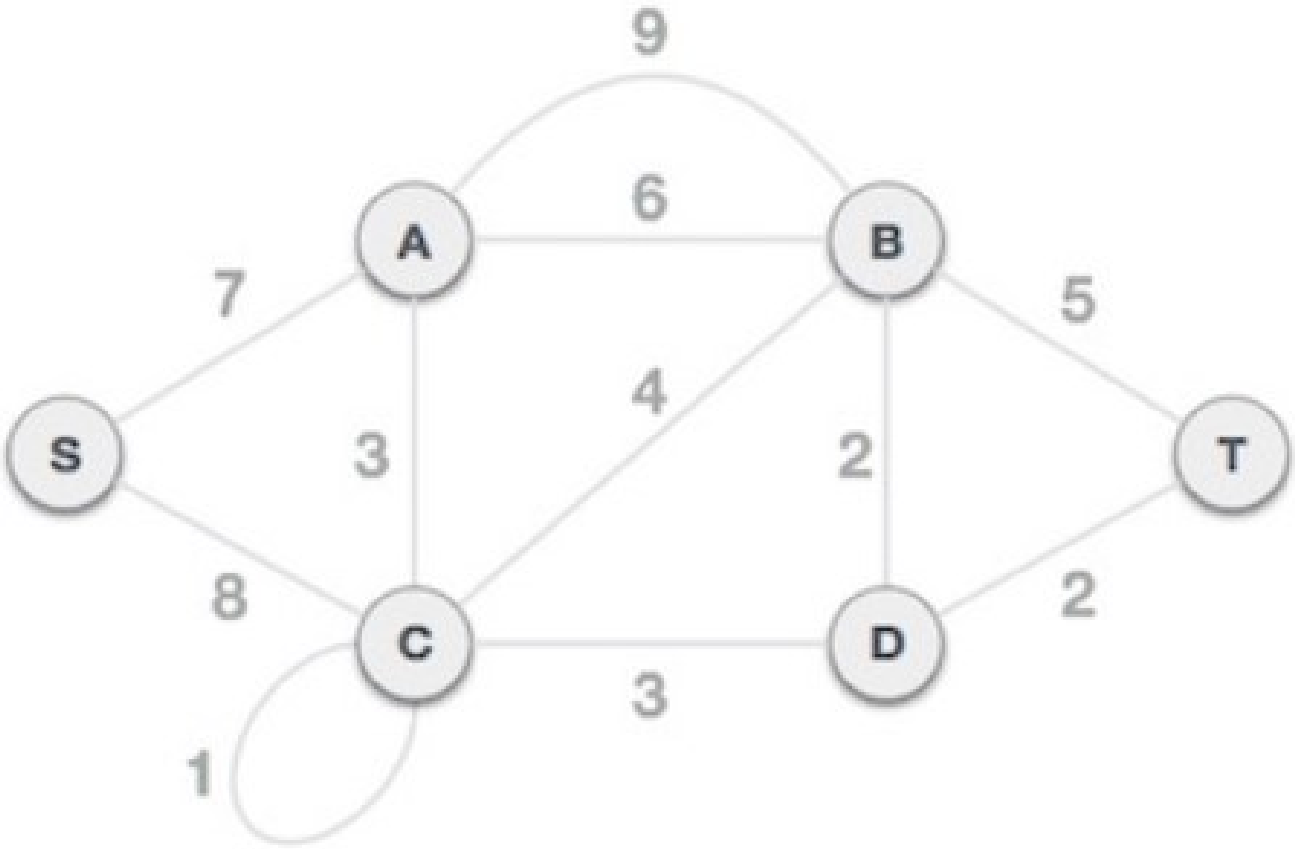
Total weight of tree: 18

Kruskal's Algorithm

Important notes:

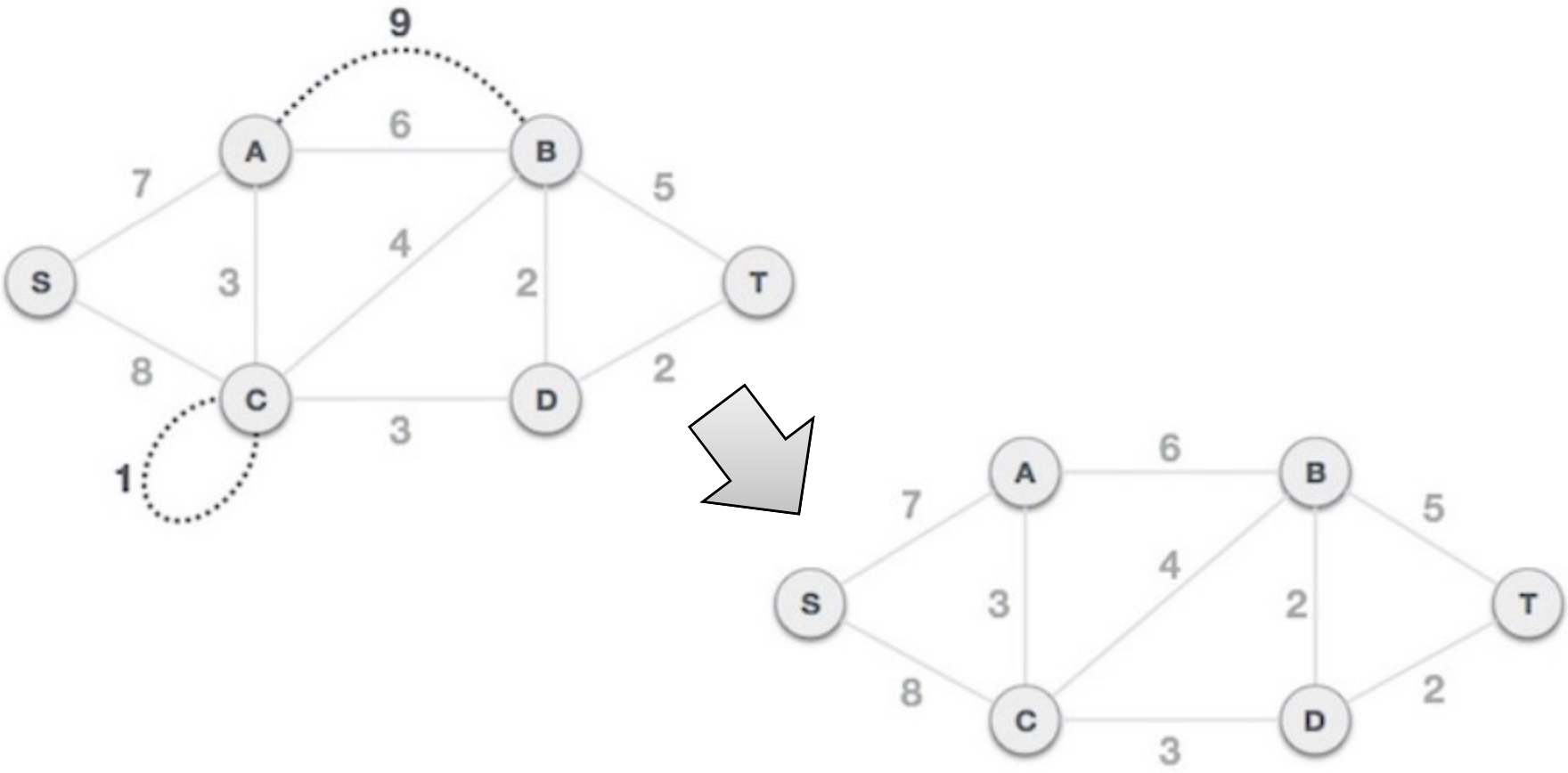
- The graph given should be a tree
  - Remove all loops (if any)
  - Remove all parallel edges
    - keep the one which has the least weight associated and remove all others

Example



Example

- Remove all loops and parallel edges



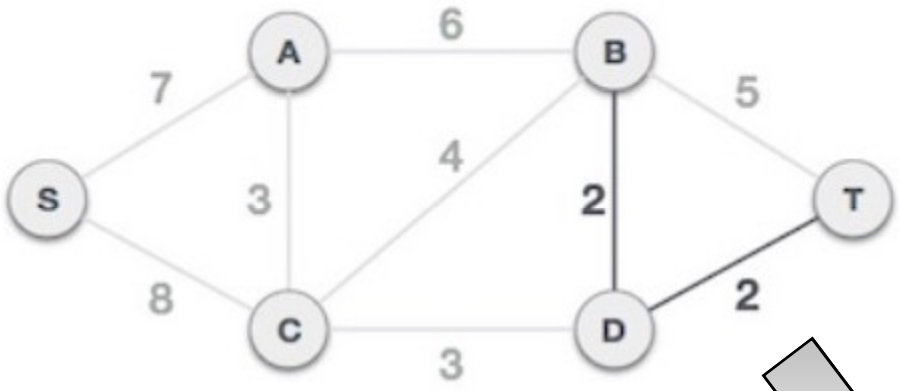
Example

- Arrange all edges in their increasing order of weight

BD	DT	AC	CD	CB	BT	AB	SA	SC
2	2	3	3	4	5	6	7	8

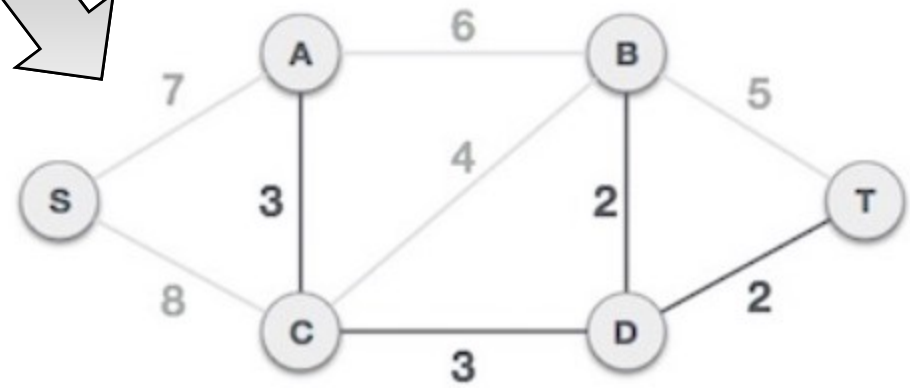
Example

- Add the edge which has the least weightage

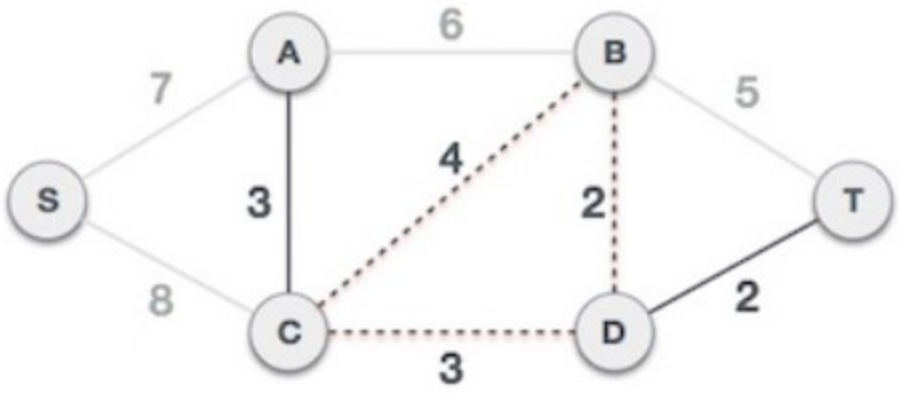


The least weight is 2 and edges involved are BD and DT

Next weight is 3, and associated edges are AC and CD

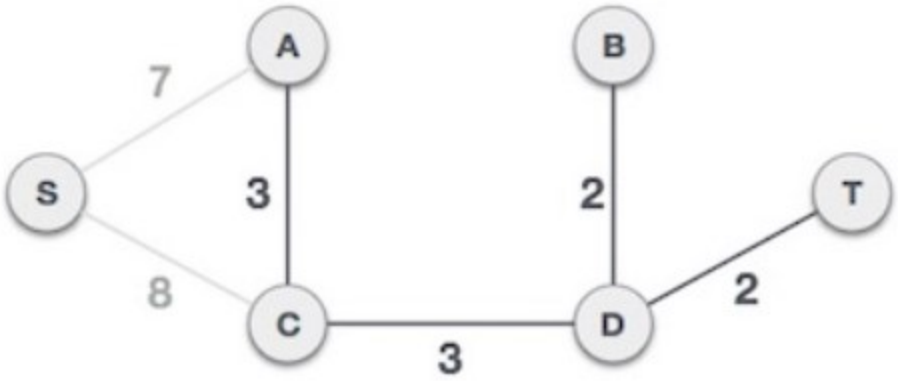


Example



Next weight is 4, and we observe that adding it will create a circuit in the graph. Thus, we ignore it.

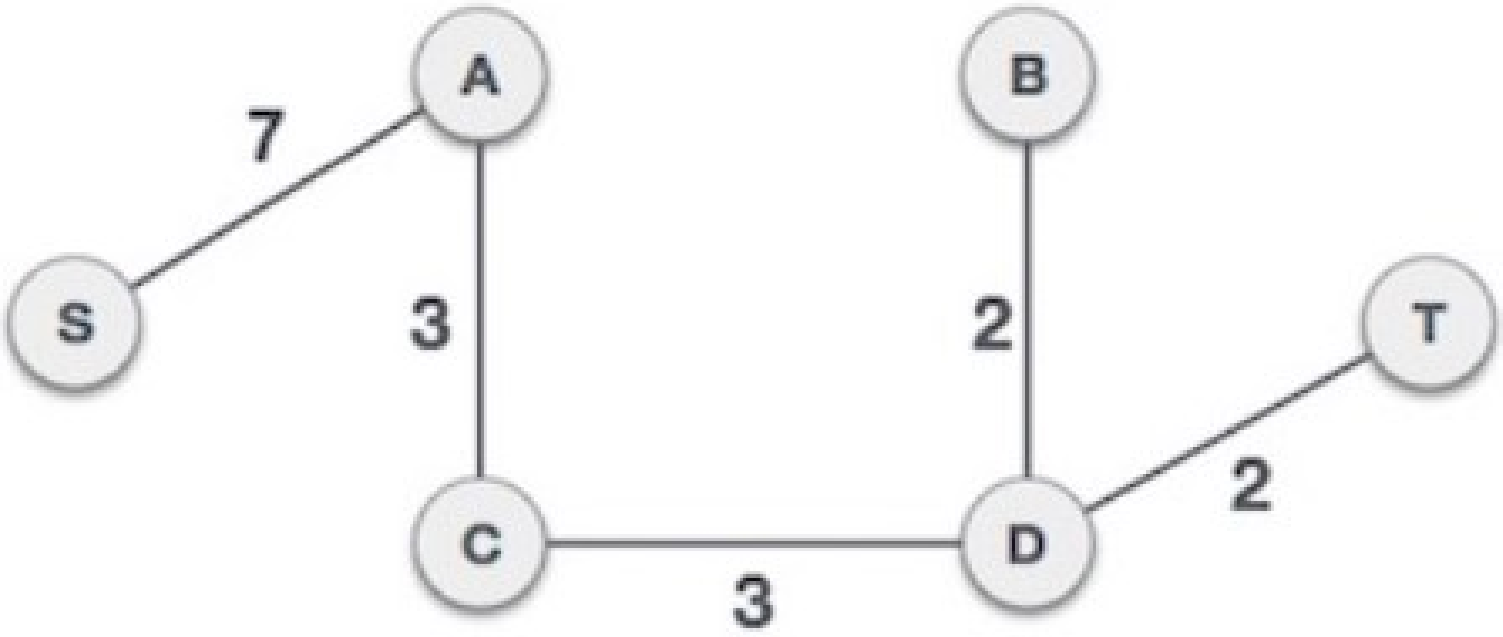
We observe that edges with weight 5 and 6 also create circuits. We ignore them and move on.



Now we are left with only one node to be added. Between the two least weighted edges available 7 and 8, we shall add the edge with weight 7.

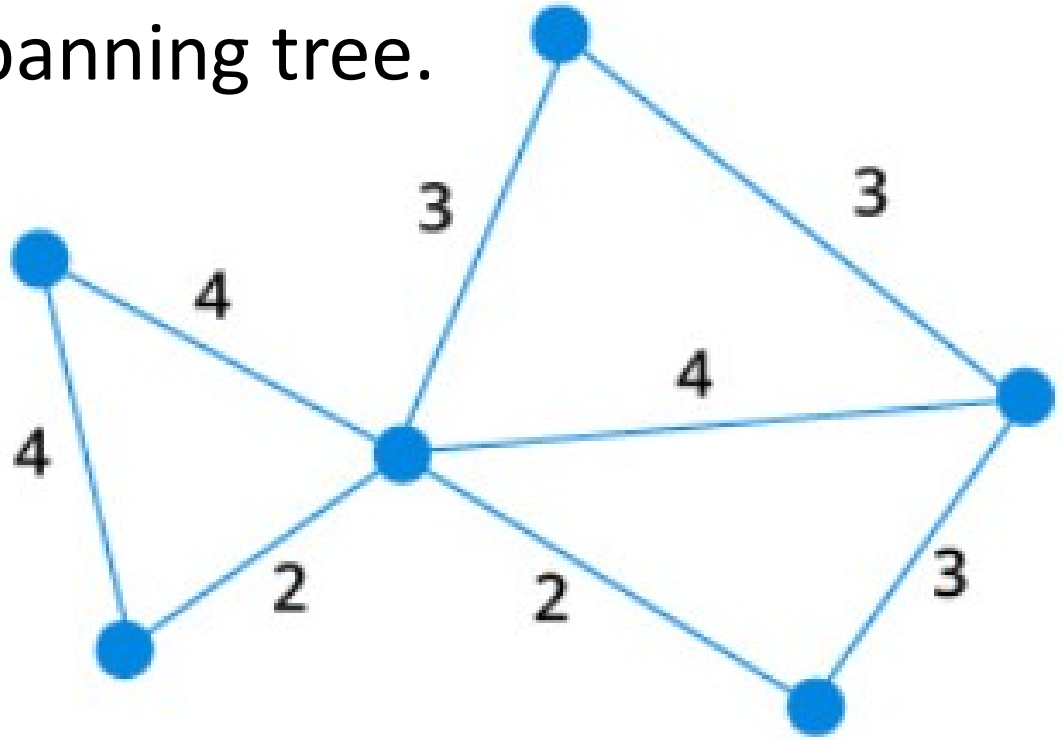
Example

Now we have minimum spanning tree with total weight is 17.



Exercise

Find the minimum spanning tree using Kraskal's Algorithm and give the total weight for the minimum spanning tree.



Exercise

Find the minimum spanning tree using Kraskal's Algorithm and give the total weight for the minimum spanning tree.

